

The Grand Unification of Mental Disease

This is a mathematical derivation, not a claim of medical truth.

Registry: [CKS-BIO-74-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-BIO-1-2026] → [CKS-BIO-73-2026] → [CKS-BIO-74-2026]

Parent Framework: [CKS-0-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological “Truth” is a category error. If the math compiles, the result is Q.E.D.

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OPERATIONAL DECLARATION

This is a mathematical derivation, not a claim of medical truth.

From CKS axioms ($D=3$, $S=2$, $\mathbb{Q}$), we derive a theoretical model where:

- **Mental states** = Registry angular positions (θ)
- **Function** = Flow efficiency $F(\theta) = F_{\text{max}} \times \cos(\theta)$
- **Pathology** = Angular deviation from 0° vertical

This framework:

- Makes specific mathematical predictions
- Requires empirical validation
- Should NOT replace evidence-based care
- Is presented for theoretical exploration only

All clinical claims require testing.

PART I: THE MATHEMATICAL FOUNDATION

§1. The Angular State Space

FROM AXIOMS D=3, S=2, $\mathbb{Q}$:

1.1 The Registry as Vector

Mathematical model:

- Registry = Vector in phase space
- Direction = Angle θ from vertical (0°)
- Magnitude = Processing capacity

Optimal state:

- $\theta = 0^\circ$ (vertical alignment)
- Points toward N=0 Pivot
- Maximum flow efficiency

Deviated states:

- $0^\circ < \theta < 180^\circ$
- Reduced flow efficiency
- Predicted functional impairment

1.2 The Flow Function

AXIOM 1.1: Angular Flow Law

$$F(\theta) = F_{\max} \times \cos(\theta)$$

Where:

- $F(\theta)$ = Processing/venting efficiency at angle θ
- $F_{\max}$ = Maximum efficiency (at $\theta = 0^\circ$)
- θ = Deviation angle from vertical

Mathematical properties:

- $F(0^\circ) = F_{\max} \times 1 = F_{\max}$ (optimal)
- $F(45^\circ) = F_{\max} \times 0.707 = 71\%$ (reduced)

- $F(90^\circ) = F_{\text{max}} \times 0 = 0$ (blocked)
- $F(180^\circ) = F_{\text{max}} \times -1 = -F_{\text{max}}$ (inverted)

This is geometric, not biological.

Pure mathematics from angular alignment.

§2. The State Classification

Mathematical partitioning of angle space:

2.1 The θ Taxonomy

THEOREM 2.1: Discrete State Zones

Zone 1: Optimal Function ($\theta = 0\text{-}15^\circ$)

- $F(\theta) \geq 0.97$
- Minimal deviation
- High coherence predicted

Zone 2: Moderate Dysfunction ($\theta = 15\text{-}60^\circ$)

- $0.50 \leq F(\theta) < 0.97$
- Partial flow reduction
- Proportional impairment predicted

Zone 3: Severe Dysfunction ($\theta = 60\text{-}90^\circ$)

- $0 \leq F(\theta) < 0.50$
- Major flow reduction
- Severe impairment predicted

Zone 4: Critical Lock ($\theta = 90^\circ$)

- $F(\theta) = 0$
- Complete flow cessation
- Total blockage predicted

Zone 5: Inversion ($\theta = 90\text{-}180^\circ$)

- $F(\theta) < 0$
- Reversed flow direction
- Paradoxical states predicted

Purely mathematical zones.

No biological claim.

Testable predictions.

2.2 Mapping to Clinical Observations

HYPOTHESIS 2.1: Angular-Clinical Correspondence

If angle determines function (from axioms),

Then clinical states should map to angles:

Proposed mapping (TESTABLE):

$\theta \approx 0-15^\circ$: “Normal” function

- Prediction: $F \geq 0.97$
- Observable: High social/cognitive function
- Test: Measure angle, correlate with function

$\theta \approx 30-45^\circ$: “Anxious/ADHD” states

- Prediction: $F \approx 0.71-0.87$
- Observable: Scattered attention, hypervigilance
- Test: Angle during anxiety vs. calm

$\theta \approx 60^\circ$: “Depressive” states

- Prediction: $F \approx 0.50$
- Observable: Reduced energy, “heaviness”
- Test: Angle correlation with mood scales

$\theta = 90^\circ$: “Locked” states

- Prediction: $F = 0$
- Observable: Severe cognitive loops, withdrawal
- Test: Angle in catatonia, severe autism

$\theta = 180^\circ$: “Inverted” states

- Prediction: $F = -1$
- Observable: Psychotic symptoms
- Test: Angle during hallucination periods

NOTE: These are PREDICTIONS, not facts.

Require empirical validation.

Could be completely wrong.

§3. The R-Accumulation Model

FROM BIO-69, BIO-70, BIO-72:

3.1 The Viscosity Hypothesis

THEOREM 3.1: Remainder-Induced Toppling

Hypothesis: R-density affects angular stability

Mathematical model:

- R = Remainder/noise accumulation
- $R_{\text{baseline}} = 19$ (from hexagonal geometry)
- $R_{\text{threshold}} = 69$ (from sovereignty calculation)

Predicted relationship:

$\theta \approx f(R)$, where f is increasing function

Proposed mechanism:

1. Low R (< 30): $\theta \approx 0^\circ$ (stable vertical)
2. Moderate R (30-50): θ increases (drift begins)
3. High R (50-69): $\theta \rightarrow 90^\circ$ (approaching topple)
4. Critical R (≥ 69): $\theta = 90^\circ$ (locked horizontal)

This predicts:

- R -reduction $\rightarrow$ angle improvement
- R -increase $\rightarrow$ angle worsening
- Threshold at $R = 69$

Testable via:

- Measure R -proxy (metabolic markers, HRV, etc.)
- Measure angle (imaging, functional tests)
- Correlate over time and interventions

3.2 Sources of R (Theoretical)

HYPOTHESIS 3.1: R-Accumulation Pathways

If R determines angle,

Then R-sources should predict dysfunction:

Predicted R-sources:

1. Sleep deprivation (no nightly clearing)
2. Chronic stress (sustained high input)
3. Inflammatory diet (high-R fuel)
4. Sedentary lifestyle (reduced venting)
5. Social isolation (reduced external flow)
6. Unprocessed trauma (stuck loops)

Each should correlate with:

- Increased R-markers
- Increased angular deviation
- Decreased function

Testable by:

- Intervention studies (modify each factor)
- Measure R and θ before/after
- Look for predicted relationships

Could be wrong. Needs testing.

§4. The Bilateral Model (S=2)

FROM AXIOMS:

4.1 The Side A/B Hypothesis

THEOREM 4.1: Bilateral Processing

From S=2 axiom:

- Two-sided manifold structure
- Side A and Side B must integrate
- Integration requires proper alignment

Mathematical prediction:

At $\theta = 0^\circ$: Sides integrate smoothly

At $\theta = 90^\circ$: Integration impaired

At $\theta = 180^\circ$: Sides inverted

Predicted phenomenology:

$\theta = 0^\circ$ (normal):

- $A + B \rightarrow$ Unified perception
- Smooth bilateral integration
- No internal conflict

$\theta = 90^\circ$ (locked):

- A and B misaligned
- Integration failure
- Possible sensory fragmentation

$\theta = 180^\circ$ (inverted):

- A and B reversed
- B perceived as external
- Possible hallucination mechanism

This is testable:

- Measure bilateral coherence (EEG, fMRI)
- Compare across angular states
- Look for predicted patterns

Pure theory. Needs validation.

§5. Specific Condition Models

Mathematical predictions for specific states:

5.1 Depression ($\theta \approx 60^\circ$)

MODEL 5.1: The Oblique Drift

Mathematical prediction:

At $\theta = 60^\circ$:

$$F(60^\circ) = \cos(60^\circ) = 0.50$$

Predicted effects:

- 50% reduction in processing capacity
- Proportional “heaviness” (inefficiency)
- Reduced qualia vividness
- Slowed temporal flow

Observable correlates (testable):

- Reduced metabolic activity (PET scan)
- Slowed reaction times
- Reduced HRV (autonomic measure)
- Reports of “heaviness,” “fog”

Intervention prediction:

If angle can be reduced ($60^\circ \rightarrow 30^\circ$):

- F increases ($0.50 \rightarrow 0.87$)
- Function should improve proportionally

Test via:

- Measure baseline angle
- Apply intervention (see §8)
- Remeasure angle and function
- Check for predicted relationship

5.2 Anxiety/ADHD ($\theta \approx 15\text{-}45^\circ$, oscillating)

MODEL 5.2: Angular Instability

Mathematical prediction:

Not fixed angle, but rapid oscillation:

$\theta(t)$ oscillates between 0° and 45°

Predicted effects:

- F varies rapidly (0.87-1.0)
- Unstable processing
- High metabolic cost (constant re-alignment)
- Hypervigilance (seeking stability)

Observable correlates (testable):

- High-frequency EEG changes
- Variable HRV
- Attention shifts
- Restlessness

This predicts:

Anxiety $\neq$ fixed state

But instability around optimal

Intervention should:

- Dampen oscillations
- Stabilize near 0°
- Reduce variance, not just mean

Testable via time-series analysis.

5.3 Psychosis ($\theta \approx 180^\circ$)

MODEL 5.3: The Side-B Crossing

Mathematical prediction:

At $\theta = 180^\circ$:

$F(180^\circ) = -1.0$ (inverted flow)

From S=2 bilateral model:

- Side B perceived as external
- Internal signals appear from “outside”
- Self/other boundary inverted

Predicted phenomenology:

- Hallucinations (Side B leakage)
- Delusions (inverted causality)
- Thought insertion (internal $\rightarrow$ external)

Observable correlates (testable):

- Bilateral EEG asymmetry
- Abnormal source localization
- Reality monitoring deficits

Intervention prediction:

Restore θ to $< 90^\circ$:

- Should reduce positive symptoms
- May not affect negative symptoms

Very speculative.

Requires careful testing.

5.4 Autism ($\theta = 90^\circ$, hard lock)

MODEL 5.4: The Metallic Pin Hypothesis

From BIO-73 heavy metal model:

Mathematical prediction:

If heavy metals create high-R anchors,

Then θ locked at 90° by gravitational well

Predicted mechanism:

- Metal atoms: $R \gg 69$
- Create local Id-layer distortion
- Pin angle at 90° (cannot escape)

$F(90^\circ) = 0$:

- No external venting possible
- Must use kinetic methods (stimming)
- Sensory integration impaired (bilateral)

Testable predictions:

1. Heavy metal correlation:
 - Measure metal levels
 - Correlate with symptom severity
 - Predict: positive correlation
2. Chelation effect:
 - Remove metals (if present)
 - Predict: angle improvement
 - Predict: symptom reduction
3. Stimming function:

- Prevents when angle = 90°
- Predict: increased internal pressure
- Allow stinging: pressure releases

NOTE: Heavy metal hypothesis controversial.

Requires rigorous testing.

May be completely wrong.

PART II: TESTABLE PREDICTIONS

§6. Falsification Criteria

How to prove this framework wrong:

6.1 Core Predictions

PREDICTION 1: Angle Measurability

- Must be able to measure θ somehow
- Via imaging, EEG, functional tests, etc.
- If angle cannot be defined/measured $\rightarrow$ framework fails

PREDICTION 2: Angle-Function Correlation

- $F(\theta) = \cos(\theta)$ must hold
- Function must correlate with angle
- If no correlation $\rightarrow$ framework fails

PREDICTION 3: Angle Modifiability

- Interventions should change θ
- Changed θ should predict changed function
- If angle fixed despite intervention $\rightarrow$ framework fails

PREDICTION 4: R-Angle Relationship

- R-reduction should improve angle
- R-increase should worsen angle
- If no relationship $\rightarrow$ framework fails

PREDICTION 5: Bilateral Integration

- S=2 effects should be measurable

- Should vary with angle
- If bilateral processing unrelated → framework fails

6.2 Specific Falsifications

Framework REJECTED if:

1. Function independent of angle
 - People function perfectly at 90°
 - Or dysfunction at 0°
 - No systematic relationship
 2. Interventions don't affect angle
 - SMR protocol changes function
 - But angle remains unchanged
 - Mechanism not geometric
 3. R irrelevant to angle
 - R varies widely
 - Angle doesn't track
 - No causal relationship
 4. No bilateral effects
 - Side A/B processing identical
 - No integration measured
 - S=2 not relevant
 5. Better model exists
 - Different framework
 - Better predictions
 - Simpler mathematics
-

§7. Measurement Proposals

How to test this framework:

7.1 Angle Measurement Methods

PROPOSAL 1: fMRI Vector Analysis

- Map neural flow patterns
- Calculate dominant directional vector
- Measure angle from “vertical” (baseline)
- Compare across conditions

PROPOSAL 2: EEG Phase Coherence

- Bilateral hemisphere synchrony
- Phase relationship analysis
- Angular representation of coherence
- Track over time

PROPOSAL 3: Functional Performance

- Cognitive task battery
- Social interaction measures
- Emotional regulation tests
- Calculate composite “flow” measure
- Infer angle from $F = \cos(\theta)$

PROPOSAL 4: HRV-Based Proxy

- Heart rate variability as R-measure
- Autonomic balance as alignment proxy
- Correlate with other measures
- Validate as angle indicator

All require validation.

May not work.

7.2 Intervention Studies

STUDY DESIGN 1: Sonic Frequency

Population: 200 subjects (depression, anxiety, controls)

Intervention: 40-60 Hz audio exposure

- 30 min, 2 × daily
- 12 weeks

Measures:

- Angle proxy (pre/post)
- Clinical scales (PHQ-9, GAD-7)
- Functional measures

Prediction: Angle $\rightarrow$ 0°, symptoms reduce

STUDY DESIGN 2: Postural Training

Population: 150 subjects (mixed conditions)

Intervention: Vertical alignment training

- Daily practice
- 8 weeks

Measures:

- Posture (objective)
- Angle proxy
- Function

Prediction: Better posture $\rightarrow$ better angle $\rightarrow$ better function

STUDY DESIGN 3: R-Reduction Protocol

Population: 100 subjects (high baseline R)

Intervention: Multi-modal R-reduction

- Sleep optimization
- Anti-inflammatory diet
- Stress reduction
- Exercise

Measures:

- R-markers (HRV, metabolic)
- Angle proxy
- Clinical outcomes

Prediction: R $\downarrow$ $\rightarrow$ θ $\downarrow$ $\rightarrow$ Function $\uparrow$

All require funding and approval.

May show no effects.

Open to negative results.

§8. The SMR Protocol (Theoretical)

Mathematical prediction of effective interventions:

8.1 Angular Restoration Methods

IF angle determines function (from axioms)

THEN methods to improve angle should improve function

Predicted effective methods:

METHOD 1: Sonic (40-60 Hz)

- Theoretical: Matches registry frequencies
- Prediction: Reduces R-viscosity
- Prediction: Allows angle improvement
- Testable: Before/after comparison

METHOD 2: Postural (0° vertical)

- Theoretical: Physical → neural mapping
- Prediction: External alignment → internal alignment
- Prediction: Trains angle toward vertical
- Testable: Posture intervention studies

METHOD 3: R-Reduction

- Theoretical: Lower R → less angular pull
- Methods: Sleep, diet, stress, exercise
- Prediction: Multi-modal superior to single
- Testable: Factorial design

METHOD 4: Cognitive (acceptance)

- Theoretical: Resistance creates R
- Prediction: Acceptance reduces R
- Prediction: Angle improves
- Testable: Mindfulness RCT

IMPORTANT: These are predictions, not proven treatments.

Current evidence-based care should continue.

These are hypotheses to test, not replace care.

8.2 Protocol Specification

Theoretical optimal protocol (UNTESTED):

Daily (prevention/maintenance):

- 10 min: 60 Hz humming
- Throughout: Vertical posture awareness
- 16:8: Eating window (R-reduction)
- 8 hrs: Dark sleep (nightly reset)

Weekly (deep work):

- 30 min: Extended sonic practice
- 60 min: Movement/exercise
- Social engagement
- Nature exposure

Monthly (reset):

- 24-48 hr: Extended fast
- Intensive alignment work
- Review and adjust

This is THEORETICAL.

Not medical advice.

Requires testing.

May be ineffective or harmful.

PART III: LIMITATIONS AND CAVEATS

§9. What This Framework Cannot Do

LIMITATION 1: Not Proven

- These are mathematical derivations
- From theoretical axioms
- No empirical validation yet
- Could be completely wrong

LIMITATION 2: Not Complete

- Doesn't explain all variance
- Other factors clearly important
- Genetics, development, etc.
- At best partial explanation

LIMITATION 3: Not Immediate Cure

- Even if correct, implementation complex
- Individual variation significant
- Time scales unclear
- No guarantees

LIMITATION 4: Not Replacement

- Current treatments work for many
- Evidence base exists
- Should not abandon proven care
- Complementary at most

LIMITATION 5: Not Simple

- Mental illness is complex
- Multiple interacting factors
- Biology, psychology, social, environmental
- Reductionism risky

§10. Research Questions

What needs to be investigated:

10.1 Fundamental Questions

Q1: Can angle be measured reliably?

- What methods work?
- What's the test-retest reliability?
- Does it correlate across methods?

Q2: Does angle correlate with function?

- Predicted $F = \cos(\theta)$ relationship?

- Across what domains?
- What's the effect size?

Q3: Can angle be modified?

- Which interventions work?
- How much change possible?
- How long does change last?

Q4: Does angle change predict outcome?

- Better than current predictors?
- Incremental validity?
- Clinical utility?

Q5: What are moderators?

- Who responds to angle interventions?
- Baseline angle?
- Chronicity?
- Other factors?

10.2 Mechanistic Questions

Q6: What causes angular deviation?

- R-accumulation (as predicted)?
- Other factors?
- Relative contributions?

Q7: What maintains deviation?

- Biological factors?
- Behavioral patterns?
- Environmental?
- Combination?

Q8: How does angle affect biology?

- Neural correlates?
- Neurotransmitter systems?
- Autonomic function?
- Immune function?

Q9: What's the temporal dynamics?

- How fast does angle change?
 - Natural history?
 - Intervention timeline?
 - Relapse patterns?
-

§11. Integration with Existing Knowledge

How this relates to current science:

11.1 Points of Contact

OVERLAP 1: Autonomic Balance

- Angle $\leftrightarrow$ Sympathetic/parasympathetic
- HRV as possible proxy
- Polyvagal theory connections
- Testable overlap

OVERLAP 2: Postural Effects

- Embodied cognition research
- Power posing studies (controversial)
- Posture-mood relationships
- Existing literature to build on

OVERLAP 3: Oscillatory Dynamics

- 40 Hz gamma and cognition
- Frequency-specific effects
- Brain stimulation research
- Some empirical basis

OVERLAP 4: Stress and Inflammation

- R as stress/inflammation proxy
- Established health effects
- Mechanistic pathways known
- Framework compatible

OVERLAP 5: Acceptance-Based Therapies

- ACT, mindfulness
- Reduce experiential avoidance
- May reduce R-accumulation
- Complementary mechanisms

11.2 Points of Divergence

DIFFERENCE 1: Geometric Primary

- Current: Chemistry/biology primary
- CKS: Geometry primary
- Major paradigm shift
- Requires strong evidence

DIFFERENCE 2: Unified Mechanism

- Current: Separate disorders
- CKS: Variations on angle theme
- Controversial claim
- Needs demonstration

DIFFERENCE 3: Simple Interventions

- Current: Complex treatments
- CKS: Geometric realignment
- Seems too simple
- Extraordinary claim needs extraordinary evidence

DIFFERENCE 4: Reincarnation/Id Layer

- Current: Not in model
- CKS: Fundamental to framework
- Major metaphysical claim
- Cannot test directly

§12. Conclusion: A Testable Framework

Final summary:

12.1 What We've Derived

From axioms $D=3$, $S=2$, $\mathbb{Q}$:

1. Angular state model
 - θ as fundamental variable
 - $F(\theta) = \cos(\theta)$ relationship
 - Discrete state zones
2. Clinical mapping
 - Depression $\approx 60^\circ$
 - Anxiety $\approx 15\text{-}45^\circ$ oscillation
 - Psychosis $\approx 180^\circ$
 - Autism $\approx 90^\circ$ (locked)
3. Intervention predictions
 - Sonic frequencies
 - Postural alignment
 - R-reduction
 - Should improve angle $\rightarrow$ function
4. Falsification criteria
 - Specific, testable predictions
 - Multiple ways to be wrong
 - Open to negative results

All mathematics.

No claims of truth.

Requires validation.

12.2 Next Steps

IMMEDIATE:

1. Develop angle measurement methods
2. Pilot test in small samples
3. Validate across measures
4. Refine predictions

SHORT-TERM:

1. Correlation studies (angle $\leftrightarrow$ function)
2. Small intervention trials
3. Mechanism exploration
4. Theory refinement

LONG-TERM:

1. Large-scale RCTs
2. Mechanism studies
3. Clinical implementation (if validated)
4. Integration with existing treatments

Or: Framework disproven early

Move to better models

Science progresses

12.3 The Value Proposition

IF validated (big if):

Advantages:

- Unified understanding
- Simple core mechanism
- Clear intervention targets
- Measurable outcomes
- Preventable conditions

IF disproven (also valuable):

Still useful:

- Generates testable hypotheses
- May find partial truths
- Eliminates wrong ideas
- Advances science

Either way: Testable framework

Science works through testing

This provides tests to run

Results will inform us

END CKS-BIO-74-2026

Status: Mathematical Derivation Complete

Empirical Status: UNTESTED

Clinical Status: NOT VALIDATED

Recommendation: RESEARCH ONLY

This framework:

- Makes specific predictions ✓
- Can be tested ✓
- Can be falsified ✓
- Should NOT replace care ✓
- Requires validation ✓

Critical points:

1. This is theory, not fact
2. Continue evidence-based care
3. Test predictions rigorously
4. Accept negative results
5. Science proceeds through testing

The mathematics is complete.

The empirical work begins.

Let testing determine truth.

Q.E.D. (Mathematics)

Q.T.B.D. (Empirical Reality)

FINAL DISCLAIMER:

This document presents mathematical derivations from theoretical axioms. Nothing herein constitutes medical advice, diagnosis, or treatment recommendation. All clinical applications require rigorous empirical validation through properly designed studies. Individuals experiencing mental health concerns should seek care from qualified healthcare professionals using evidence-based treatments. The author makes no claims regarding the truth or efficacy of any intervention discussed. This is a theoretical framework for scientific investigation, not a clinical protocol.

[**CKS-BIO-74-2026**] The Grand Unification of Mental Disease. Github: <https://github.com/ghowland/cks/tree/main/papers/BIO-74-2026>

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-0-2026>

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026>

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[**CKS-BIO-73-2026**] The 90-Degree Phase Lock. Github: <https://github.com/ghowland/cks/tree/main/papers/BIO/CKS-BIO-73-2026>